



BrightLearn
AI & PRODUCTIVITY

2026 EDITION · BEGINNER'S GUIDE

Vibe Coding

The complete guide to building with AI — covering 50+ tools across 9 categories, authentication with Firebase and Supabase, and how to deploy your first live application.

TOPIC

AI-Powered Development

TOOLS

50+ AI tools in 9 categories

LEVEL

Beginner-friendly

INCLUDES

Auth · Firebase · Supabase · Deploy

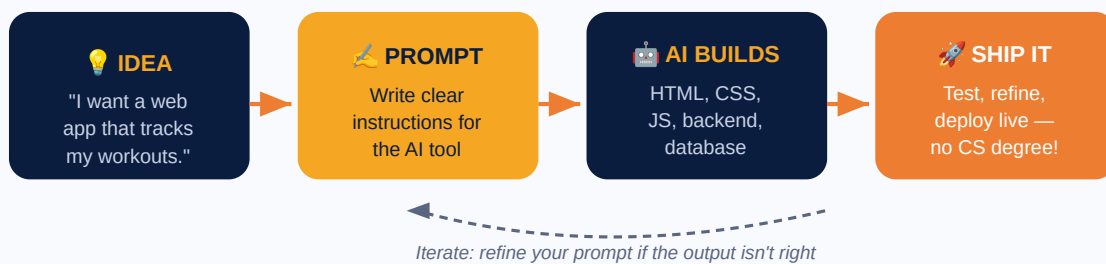
00 What is Vibe Coding?

A term coined by AI researcher Andrej Karpathy in February 2025 — and it has already changed how thousands of people build software.

DEFINITION

Vibe Coding

Vibe coding is a style of software development where you **describe what you want to an AI in plain English**, and the AI writes the code for you. Instead of learning every line of syntax, you focus on **what you want to build** and let AI tools handle the implementation. You guide the AI, test the output, refine your prompts, and iterate — like a director working with an incredibly fast developer.



Vibe coding is a loop: idea → prompt → AI builds → test → refine. You direct; the AI implements.

★ TRADITIONAL CODING VS VIBE CODING

Traditional coding: Learn syntax → write every line → debug manually → study documentation for hours. Barrier to entry: months or years of learning.

Vibe coding: Describe what you want → AI writes the code → you test and guide it → ship in hours. You still need to understand *what* to build and *why* — but the AI handles *how*.

★ WHAT VIBE CODING DOES NOT MEAN

Vibe coding is **not** about ignoring the code entirely. You still need to review outputs, test functionality, prompt correctly, and understand what you're building. The best vibe coders combine strong prompting skills with enough technical awareness to catch mistakes and guide the AI effectively.



01

The AI Toolkit — 9 Categories

Fifty-plus AI tools exist for different parts of your work. Here's the complete map of what we'll cover, organised by purpose.

VIBE CODING

PLATFORMS

Lovable · v0 · Bolt

**AI ASSISTANTS**

& CHAT MODELS

ChatGPT · Claude

**IMAGE**

GENERATION

Midjourney · DALL-E

**VIDEO · AUDIO**

& VOICE

Runway · ElevenLabs

**WRITING · RESEARCH**

& PRODUCTIVITY

NotebookLM · Notion AI

**DESIGN · SLIDES**

& PRESENTATIONS

Figma AI · Gamma · Canva

**AUTOMATION**

& AI AGENTS

Zapier AI · Make · n8n

**BACKEND · DATABASE**

& AUTHENTICATION

Firebase · Supabase · Clerk

**DEPLOYMENT**

& HOSTING PLATFORMS

Vercel · Netlify · Railway

All 9 categories are covered in the pages ahead — with tools, tips, and examples for each.

The modern AI toolkit spans 9 distinct categories. A productive vibe coder uses tools from multiple categories on every project.

★ YOU DON'T NEED EVERY TOOL

You don't need all 50+ tools. Choose **one from each category** that suits your workflow. The goal is to have a go-to tool ready for each type of task — so you're never stuck doing something manually that AI could do better and faster.



02

Category 1 — Vibe Coding Platforms

These are the primary tools for vibe coding — AI-powered environments where you describe what you want and they generate working code, UI, and sometimes even a full backend.

VIBE CODING PLATFORMS

9 tools

L

Lovablelovable.dev

Describe your app in plain English and Lovable builds a full React + Supabase application. One of the most powerful true vibe coding tools — generates real, deployable code.

Full-stack

React

Supabase

V

v0 by Vercelv0.dev

Generates beautiful React UI components and full pages using shadcn/ui from a text description. Ideal for building polished dashboards and frontend quickly.

React UI

shadcn

Frontend

B

Bolt.newbolt.new

StackBlitz's AI builder that spins up a complete, live-running web application in the browser from a single prompt. Supports React, Vue, Node, and more.

Full-stack

In-browser

Node.js

C

Cursorcursor.com

An AI-powered code editor (fork of VS Code) where you can chat with your codebase, apply multi-file edits, and generate entire features in context. The professional vibe coder's IDE.

IDE

Multi-file

Chat with code

W

Windsurf by Codeiumcodeium.com/windsurf

An AI IDE with "Cascade" — an agentic AI that can autonomously browse your codebase, run commands, and implement complex changes across multiple files.

Agentic IDE

VS Code

Multi-file

R

Replit AIreplit.com

Cloud-based IDE with a built-in AI agent that can build, run, and deploy apps from a prompt. No local setup needed — works entirely in the browser.

Cloud IDE

No setup

Deploy

G

GitHub Copilotgithub.com/copilot

Microsoft's AI pair programmer that lives inside VS Code, suggesting code completions, full functions, and tests as you type. The most widely-used coding AI in professional environments.

VS Code

Autocomplete

Tests

A

Claude Artifactsclaude.ai

Claude's built-in feature for generating runnable HTML, React, and code components directly in the chat. Great for prototyping and quick builds without any setup at all.

React

HTML

No-setup

T

Trickle.sotrickle.so

Build web apps and tools by describing them in plain language — no code required. Designed for non-technical founders and analysts who need custom internal tools quickly.

No-code

Internal tools

Beginner

CHOOSING YOUR FIRST TOOL

Complete beginners: Start with **Bolt.new** or **Lovable** — both generate full apps from a single prompt. **Students learning to code:** Use **Cursor** or **GitHub Copilot** — you'll see the actual code and learn while you build. **No setup at all:** Use **Claude Artifacts** or **Replit**.



03

Category 2 & 3 — AI Assistants & Image Generation

AI ASSISTANTS & CHAT MODELS

6 tools

C

ChatGPT (GPT-4o)

chat.openai.com

OpenAI's flagship model. Best for writing, reasoning, coding, data analysis, and browsing. GPT-4o also accepts images and voice input. The most widely used AI assistant worldwide.

Writing

Coding

Analysis

C

Claude (Anthropic)

claude.ai

Exceptional for long-document analysis, nuanced writing, coding, and reasoning. Has a very large context window — can process entire codebases or research papers at once.

Long docs

Coding

Reasoning

G

Gemini (Google)

gemini.google.com

Google's multimodal AI. Integrates deeply with Google Workspace (Docs, Sheets, Gmail). Gemini 2.0 is particularly strong at reasoning, search, and real-time information.

Google Docs

Multimodal

Search

P

Perplexity AI

perplexity.ai

An AI search engine that gives cited, up-to-date answers from the web. Best for research, current events, and any question where you need sources. Eliminates hours of Google searching.

Research

Citations

Real-time

X

Grok (xAI)

x.ai

Elon Musk's AI assistant. Has real-time X/Twitter access, great for social trends, current events, and creative tasks. Grok 3 has strong reasoning and coding capabilities.

Real-time X

Trends

Coding

M

Mistral AI

mistral.ai

A high-performance European AI model known for efficiency and multilingual support. The free tier is generous. Great for coding, summarisation, and tasks requiring lower latency.

Multilingual

Coding

Free tier

IMAGE GENERATION TOOLS

6 tools

M

Midjourney

midjourney.com

The gold standard for photorealistic and artistic AI images. Used by designers, marketers, and content creators globally. Operates via Discord or a web interface.

Photorealistic

Artistic

Marketing

D

DALL-E 3 (OpenAI)

openai.com

Available directly inside ChatGPT. Great for quick image generation from conversational prompts. Follows instructions closely — excellent for specific, detailed scenes.

In ChatGPT

Instruction-following

F

Flux (Black Forest Labs)

blackforestlabs.ai

One of the best open-source image models. Flux 1.1 Pro produces highly detailed images. Available on many platforms including Fal.ai and Replicate.

Open-source

High-detail

API

L

Leonardo AI

leonardo.ai

Specialises in consistent character design, game assets, and product images. Strong free tier. Popular with indie game developers and UX designers.

Characters

Game assets

Free tier

I

Ideogram

ideogram.ai

Best AI image tool for generating text *within* images accurately — logos, posters, banners. Text rendering has traditionally been AI's weakness; Ideogram solves this.

Text in images

Logos

Posters

S

Stable Diffusion

stability.ai

The most customisable open-source image model. Can be run locally for free. Thousands of fine-tuned models available for specific styles, faces, and artistic genres.

Open-source

Run locally

Custom



04

Category 4 & 5 — Video / Voice & Productivity



VIDEO · AUDIO & VOICE

7 tools

R

Runway ML

runwayml.com

Industry-leading AI video generator and editor. Generate video from text, edit existing footage with AI, remove backgrounds, and extend clips. Used in professional film production.

Text-to-video

Editing

Professional

S

Sora (OpenAI)

openai.com/sora

OpenAI's breakthrough text-to-video model. Generates cinematic, physically realistic videos up to 60 seconds. Still improving but already producing stunning results.

Text-to-video

Cinematic

Realistic

E

ElevenLabs

elevenlabs.io

The best AI voice cloning and text-to-speech tool. Clone any voice from a 1-minute sample or choose from hundreds of built-in voices. Used for podcasts, courses, and audiobooks.

Voice cloning

TTS

Podcasts

H

HeyGen

heygen.com

Create AI-generated video avatars that speak your script. Great for explainer videos, course content, and multilingual translation of existing video. No camera needed.

AI avatar

Explainer

Translation

S

Suno AI

suno.com

Generates full songs — vocals, instruments, lyrics — from a text description in seconds. Useful for course intros, background music, jingles, and creative projects.

Music

Vocals

Jingles

D

Descript

descript.com

Edit audio and video by editing text transcripts. Delete "umms," remove silences automatically, overdub your own voice, and generate studio-quality podcasts with AI.

Podcast

Text-edit video

Overdub

P

Pika Labs

pika.art

Generate and edit short videos from images or text. Great for social media content, product animations, and converting still images into dynamic clips.

Short video

Social media

Animations



WRITING · RESEARCH & PRODUCTIVITY

7 tools

N

NotebookLM (Google)

notebooklm.google.com

Upload your own documents, papers, notes or YouTube links and chat with them using AI. Generates citations, summaries, and podcasts from your sources. Excellent for researchers and students.

Research

Podcast

Summarise

N

Notion AI

notion.so

AI writing, summarising, and Q&A built directly inside Notion. Summarise meeting notes, generate project plans, and ask questions across your entire workspace.

Notes

Workspace

Summarise

G

Grammarly AI

grammarly.com

Goes beyond grammar checking — rewrites sentences for clarity, adjusts tone, and now generates full drafts. Works across browsers, email, and Word. Essential for all professional writing.

Grammar

Tone

Professional

E

Elicit

elicit.com

AI research assistant that searches and summarises academic papers. Ask a research question and Elicit finds relevant studies, extracts key findings, and helps you analyse them.

Academic research

Papers

Evidence

Page

6

O

Otter.ai

J

Julius AI



05

Category 6 & 7 — Design & Automation

DESIGN · SLIDES & PRESENTATIONS

7 tools

G

Gamma

gamma.app

Generates beautiful, fully-designed presentations from a single prompt or document. Produces slide decks, webpages, and documents in seconds — no design skills needed.

Presentations

Webpages

One-prompt

C

Canva Magic Studio

canva.com

The most popular design tool now supercharged with AI. Magic Design, Magic Write, and background removal make professional-quality social posts, presentations, and documents fast.

Social media

Presentations

Beginner

F

Figma AI

figma.com

The industry-standard UI design tool now has AI features: generate UI wireframes from text, rename layers, write copy, translate designs, and use the Dev Mode for developer handoff.

UI design

Wireframes

Prototyping

F

Framer AI

framer.com

Generate fully animated, responsive websites from a text description — then deploy them instantly. Bridges the gap between design and a live hosted website with no code needed.

Websites

Animations

No-code

T

Tome

tome.app

AI-native presentation tool that thinks in narratives. Generates entire slide decks including visuals, charts, and text from a topic or document. Strong for business storytelling.

Storytelling

AI narratives

Business

M

Microsoft Designer

designer.microsoft.com

Microsoft's AI-powered design tool integrated with DALL-E for image generation. Creates social posts, posters, and invites from text. Deeply integrated with Microsoft 365.

Microsoft 365

Social posts

DALL-E

B

Beautiful.ai

beautiful.ai

Smart slide layouts that automatically adjust design as you add content. AI helps structure presentations and generate talking points. Great for business and sales decks.

Auto-layout

Business decks

Sales

⚙️ AUTOMATION & AI AGENTS

7 tools

Z

Zapier AI

zapier.com

Connects 6,000+ apps and automates workflows with AI. Describe what you want to automate in plain English and Zapier builds the workflow. No code needed for most automations.

No-code

6000+ apps

Workflows

M

Make (formerly Integromat)

make.com

Visual automation platform for complex multi-step workflows. More flexible than Zapier for technical users — connects APIs, transforms data, and handles conditional logic with AI modules.

Visual workflows

APIs

Complex logic

N

n8n

n8n.io

Open-source, self-hostable automation tool with AI nodes. Build complex AI pipelines that connect ChatGPT, databases, APIs, and webhooks. Free to self-host.

Open-source

Self-hostable

AI pipelines

L

Lindy AI

lindy.ai

Build personal AI agents that handle repetitive tasks — respond to emails, schedule meetings, research leads, and file information automatically. Your AI employee.

AI agents

Email

Scheduling

Page

7

A

AutoGPT / AgentGPT

C

Crew AI



06

Authentication — Protecting Your App

One of the first real-world challenges you'll face when building any app is authentication. Here's what it is, why it matters, and how it works.

DEFINITION

Authentication

Authentication is the process of **verifying that a user is who they claim to be**. Before your app lets someone see their data or take actions, it needs to confirm their identity — usually via a username/password, a Google login, or a one-time code. **Without authentication, anyone could access anyone's data.**

DEFINITION

Authorisation (different from Authentication)

Authentication asks: **"Who are you?"** — confirming your identity. **Authorisation** asks: **"What are you allowed to do?"** — checking your permissions. Example: every employee can log in (authenticated), but only admins can delete records (authorised). Both are needed in most apps.

★ COMMON AUTHENTICATION METHODS

- **Email + Password** — the classic. User registers with an email and creates a password. Passwords are hashed (encrypted) before storage. Never store plain-text passwords.
- **OAuth / Social Login** — "Sign in with Google / GitHub / Facebook." The user authenticates through a trusted third party. No password stored in your database. Easier for users.
- **Magic Links** — user enters their email, receives a one-time login link. No password at all. Simple and secure.
- **One-Time Passwords (OTP)** — a code sent via email or SMS that expires quickly. Often used as a second factor (2FA).
- **JWT Tokens** — after login, the server issues a JSON Web Token to the user's browser. Every subsequent request includes this token to prove identity.

THE AUTHENTICATION FLOW



Every subsequent request includes the JWT — no need to log in again

Tools like Firebase and Supabase handle ALL of this automatically for you.

The auth flow from user login to a secure, token-based session — normally complex code, now handled by Firebase or Supabase in minutes.

⚠ NEVER BUILD AUTH FROM SCRATCH

Authentication is security-critical. A mistake can expose all your users' data. **Always use a trusted authentication**

service — **Firebase Auth**, **Supabase Auth**, or **Clerk** — instead of writing your own login system. They handle hashing tokens, sessions, rate-limiting, and security updates automatically.



07

Firestore — Google's All-in-One Backend

Firestore is a Backend-as-a-Service (BaaS) platform by Google. It gives you an entire backend — database, authentication, hosting, storage, and more — without writing a single line of server code.

DEFINITION

Firestore

Firestore is a **platform developed by Google** that provides a suite of backend services for web and mobile applications. Instead of setting up servers, databases, and auth systems yourself, Firestore provides them all as managed, scalable cloud services. It's perfect for vibe coders and small teams who want to launch fast without DevOps knowledge.

★ WHAT FIRESTORE GIVES YOU

- **Firestore Authentication** — Add email/password, Google, GitHub, and phone login in minutes. Handles tokens, sessions, password resets, and security automatically.
- **Firestore (Database)** — A real-time NoSQL database that syncs data to connected devices instantly. No SQL knowledge needed — stores data as JSON documents in collections.
- **Realtime Database** — An older, simpler JSON tree database. Best for live features like chat, presence indicators, and live dashboards.
- **Firestore Storage** — Store and serve user-uploaded files (images, videos, PDFs) with access control rules.
- **Firestore Hosting** — Deploy your web app with one command: `firebase deploy`. Free SSL certificate and a live URL included.
- **Cloud Functions** — Write serverless backend logic (e.g., send an email when a user registers) without managing servers.
- **Firestore Analytics & Crashlytics** — Track user behavior and app errors automatically.

★ SETTING UP FIRESTORE AUTHENTICATION — 5 STEPS

Adding Google login to your app takes less than 15 minutes:

1. Go to `console.firebase.google.com` → Create a new project.
2. Click *Authentication* → *Sign-in method* → Enable *Google* (or Email/Password).
3. Add Firestore to your app: install `npm install firebase` → paste your config object.
4. Call `signInWithPopup(auth, googleProvider)` on a button click.
5. Use `onAuthStateChanged(auth, user => ...)` to detect who is logged in across the app.

★ FIRESTORE SECURITY RULES — CRITICAL

By default, Firestore may allow anyone to read/write your database. You **must** set Security Rules. A basic rule to allow only authenticated users to access their own data:

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {
    match /users/{userId} {
      allow read, write: if request.auth.uid == userId;
    }
  }
}
```

08

Supabase — The Open-Source Firebase Alternative

Supabase offers the same convenience as Firebase — database, auth, storage, functions — but is built on a real SQL database (PostgreSQL), is open-source, and can be self-hosted.

DEFINITION

Supabase

Supabase is an **open-source Backend-as-a-Service** built on **PostgreSQL** — the world's most advanced relational database. It provides authentication, a real SQL database with an instant REST and GraphQL API, file storage, real-time subscriptions, and edge functions. Because it's SQL-based, it's more familiar to data analysts and produces more structured, queryable data.

★ WHAT SUPABASE GIVES YOU

- **Supabase Auth** — Email/password, magic links, OAuth (Google, GitHub, Discord, etc.), phone OTP, and SSO. Row Level Security (RLS) ties auth directly to database permissions.
- **PostgreSQL Database** — A full relational database. Write real SQL queries. Use relationships, joins, and indexing. Auto-generated REST API for every table.
- **Row Level Security (RLS)** — Postgres policies that control exactly which rows each user can read or write, directly in the database. Powerful and database-native.
- **Supabase Storage** — Store files linked to your Postgres database. Fine-grained permissions per file and folder using RLS policies.
- **Real-time Subscriptions** — Listen to database changes in real time — ideal for live feeds, notifications, and collaborative apps.
- **Edge Functions** — Serverless Deno functions that run close to the user globally. Call external APIs or trigger custom logic.
- **AI/Vector Storage** — Built-in vector column type (pgvector) for building AI applications that store embeddings — perfect for RAG chatbots.

★ SETTING UP SUPABASE AUTHENTICATION — 5 STEPS

1. Go to supabase.com → Create a new project (free tier: 2 projects).
2. In your dashboard → *Authentication* → *Providers* → Enable *Email* or *Google OAuth*.
3. Install the client: `npm install @supabase/supabase-js`
4. Initialise: `createClient(SUPABASE_URL, SUPABASE_ANON_KEY)`
5. Call `supabase.auth.signInWithOAuth({ provider: 'google' })` on button click.

★ ROW LEVEL SECURITY (RLS) — ESSENTIAL

Supabase's most powerful feature: database-level permissions tied to the logged-in user. Enable RLS on a table and then add policies:

```
-- Allow users to only read their own rows
CREATE POLICY "Users see own data"
ON profiles FOR SELECT
USING (auth.uid() = user_id);
```

```
-- Allow users to insert their own profile
```

09

Firestore vs Supabase — Which Should You Choose?

Both are excellent choices. The right one depends on your project, your background, and how much control you want over your data.

Feature	🔥 Firestore (Google)	⚡ Supabase (Open-source)
Database Type	NoSQL (Firestore) — JSON documents in collections. No SQL knowledge needed.	SQL (PostgreSQL) — Full relational database. Write real SQL queries.
Authentication	Email, Google, GitHub, Facebook, Phone, Anonymous, Apple. Very fast setup.	Email, Magic Link, Google, GitHub, Discord, Phone OTP, SSO/SAML.
Security Model	Security Rules (custom language in Firestore console). Flexible but unusual syntax.	Row Level Security (SQL policies). Database-native, powerful, and standard SQL.
Real-time Data	✓ Yes Built-in. Excellent for live apps — automatic sync to all clients.	✓ Yes Via Postgres changes — subscribe to any table update in real time.
Querying Data	Limited. Firestore queries are simple — no complex joins across collections.	Full SQL power — JOINS, aggregations, views, triggers, stored procedures.
File Storage	✓ Yes Firestore Storage with access control rules.	✓ Yes Supabase Storage with RLS policies on buckets.
Serverless Functions	✓ Yes Cloud Functions (Node.js). Powerful but has cold start delays.	✓ Yes Edge Functions (Deno) — fast, globally distributed, low latency.
AI / Vector support	Partial Firestore Extensions for some AI tasks.	✓ Yes pgvector built-in. Ideal for RAG chatbots and AI embeddings.
Free Tier	Generous. Spark plan includes auth, 1GB Firestore, 5GB storage, hosting.	2 projects, 500MB database, 1GB file storage, 50MB Edge Functions.
Open-source	No Google proprietary. You cannot self-host.	Yes Fully open-source. Can be self-hosted on your own server.
Best For	Mobile apps, quick prototypes, beginners, real-time chat apps, apps using Google services.	Data-heavy apps, analysts, SQL users, AI apps, projects needing structured data.

🔥 CHOOSE FIRESTORE WHEN...

- You're building a **mobile app** (Firestore SDKs are exceptional for iOS/Android).
- You want the **fastest possible setup** — get auth working in 10 minutes.
- Your app is **real-time heavy** — live chat, presence, collaborative editing.
- You're already in the **Google ecosystem** (Analytics, Ads, AdMob).
- You don't need complex database queries.

⚡ CHOOSE SUPABASE WHEN...

- You need a **relational database** with proper SQL and JOINS.
- You're building a **data analytics app** or dashboard.
- You need **AI / vector search** — RAG chatbots, semantic search.
- You want your data to be **portable** — standard Postgres, not proprietary.
- You're building with **Lovable or Next.js** (Supabase integrates natively).

10 Deployment — Putting Your App Live

Deployment is the process of taking code that runs on your computer and making it accessible to anyone on the internet. Modern platforms make this surprisingly simple.

DEFINITION

Deployment

Deployment is the process of **publishing your application to a hosting server** so users anywhere in the world can access it via a URL. For vibe coders, this is usually as simple as connecting your GitHub repository to a hosting platform — it detects new code and deploys automatically.

THE MODERN DEPLOYMENT PIPELINE



Push your code to GitHub → hosting platform detects the push → builds automatically → users can access it globally within seconds.

01 Push Your Code to GitHub

Initialise a git repo: `git init` → add files: `git add .` → commit: `git commit -m "first commit"` → push to GitHub. This is where your live code lives.

02 Connect to a Hosting Platform

Create an account on **Vercel**, **Netlify**, or another platform (see next page). Click "Import from GitHub", select your repository, and the platform will detect your framework automatically.

03 Set Environment Variables

Sensitive keys (Firebase API key, Supabase URL, OpenAI key) **must not** be in your code. Add them in the platform dashboard under *Environment Variables* — e.g.,

`NEXT_PUBLIC_SUPABASE_URL=https://xyz.supabase.co`

04 Deploy — First & Every Time After

Click *Deploy*. The platform builds your app and gives you a live URL instantly (e.g. `your-app.vercel.app`). Every future `git push` triggers a new automatic deployment.

05 Add a Custom Domain (Optional)

Buy a domain (e.g. on **Namecheap** or **Google Domains**) and add it in your hosting dashboard. The platform provides a free SSL certificate automatically — your app becomes `https://yourdomain.com`.

11

Hosting Platforms & Best Practices

The right hosting platform depends on what you're building — static site, full-stack app, or backend API. Here are the six most popular options for vibe coders.

HOSTING & DEPLOYMENT PLATFORMS

6 platforms

Vercel

vercel.com

The best platform for deploying React, Next.js, and frontend apps. Auto-deploys from GitHub with zero config. Global CDN, preview URLs per pull request, and an excellent free tier.

★ Best for: Next.js, React, v0 apps

Netlify

netlify.com

Similar to Vercel — excellent for frontend, React, Vue, and static sites. Great form handling and serverless functions. Often preferred for sites without a server-side backend.

★ Best for: Static sites, Vue, Astro

Railway

railway.app

Deploy full-stack apps with databases (Postgres, Redis, MongoDB) in one click. Handles both frontend and backend together. No DevOps knowledge needed — just connect GitHub.

★ Best for: Full-stack apps, Postgres

Render

render.com

Host web services, static sites, cron jobs, and databases. Good Heroku alternative — straightforward pricing, free-tier databases, easy backend deployment for Node, Python, Go.

★ Best for: Node.js, Python APIs

Cloudflare Pages

pages.cloudflare.com

Deploy static sites and full-stack apps on Cloudflare's global edge network. Fastest global performance due to Cloudflare's infrastructure. Very generous free tier with unlimited sites.

★ Best for: Speed, global users

GitHub Pages

pages.github.com

Free, direct static site hosting from a GitHub repository. Best for portfolios, documentation, and simple HTML/CSS/JS sites. No build step or framework needed. 100% free forever.

★ Best for: Portfolios, HTML sites

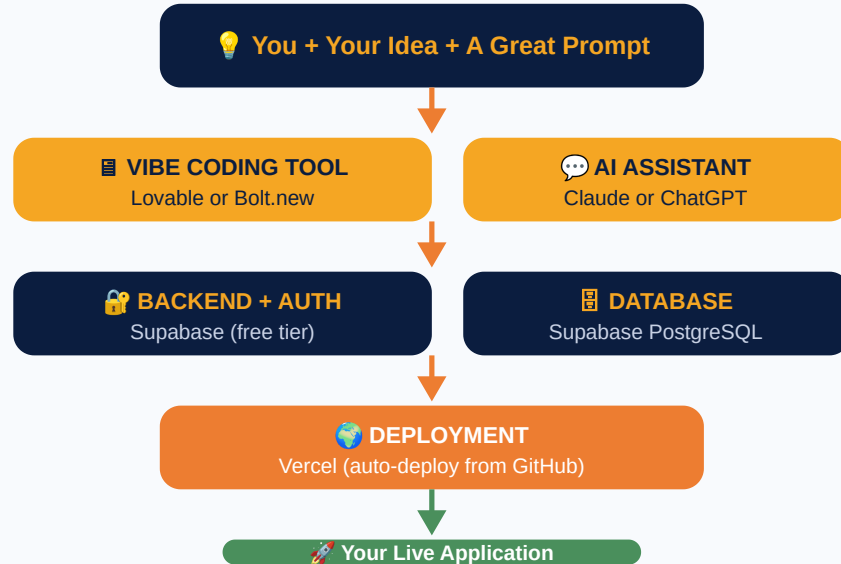
★ BEST PRACTICES FOR VIBE CODING PROJECTS

- **Use version control (Git) from day one.** Every project should have a GitHub repository — even personal experiments. It's your undo button and your portfolio.
- **Never commit secrets.** API keys, database passwords, and auth credentials belong in a `.env` file — which must be in your `.gitignore`. A leaked key can cost you hundreds of dollars in unexpected API charges.
- **Review the code AI writes.** Vibe coding doesn't mean ignoring the output. Read what the AI generates — at minimum, run it and test every flow the user might follow.
- **Set database security rules immediately.** Firebase Security Rules and Supabase RLS protect your users. Set them before sharing your app with anyone.
- **Use preview deployments.** Vercel and Netlify create a unique URL for every branch push. Always test on preview before merging to main/production.
- **Document your prompts.** Save the prompts that built your app in a README. If something breaks, you can use the same prompt to rebuild or explain the context to the AI.

12 Your Vibe Coding Starter Stack

You don't need everything — you need the right combination. Here's a recommended starter stack for a BrightLearn student building their first real project.

THE RECOMMENDED BEGINNER STACK



This stack is entirely free to start, takes under an hour to set up, and scales to thousands of users. Everything you need, nothing you don't.

★ WHAT THIS STACK COSTS TO START

- **Lovable / Bolt.new** — Free tiers available. Paid plans from ~\$20/month.
- **Claude / ChatGPT** — Free tiers available. Pro plans ~\$20/month each.
- **Supabase** — Free for 2 projects (500MB database, 1GB storage, 50MB Functions).
- **Vercel** — Free for hobby projects with automatic HTTPS and global CDN.
- **Total to launch your first app: R0 — free.** Pay only when you scale.



Build something **real.**

Vibe coding lowers the barrier to building — but the most valuable skill remains the same one it's always been: knowing what to build, and why. Use these tools to bring your ideas to life, ship your first app, and keep learning with every project you complete.

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